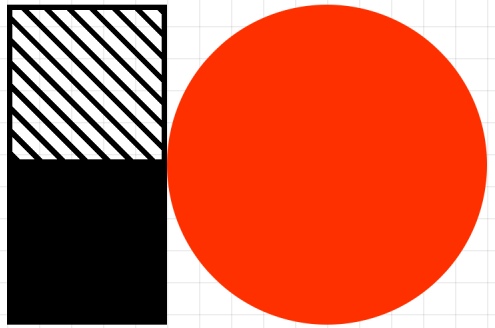


THE TECHNOLOGY VALUE STREAM

Lead time, processing time, and the road from months to minutes

FERNANDO CONTRERAS BELLEVUE UNIVERSITY OCTOBER 2026

WHAT IS A VALUE STREAM?

IN MANUFACTURING

"The sequence of activities an organization undertakes to deliver upon a customer request."

Martin and Osterling, quoted in Kim et al. (2021)

IN DEVOPS

The process required to convert a business hypothesis into a technology-enabled service or feature that delivers value to the customer.

Starts when work is accepted into Development and added to the committed backlog

VALUE IS CREATED ONLY WHEN A SERVICE IS RUNNING IN PRODUCTION.

Source: Kim et al. (2021), ch. 1

DEPLOYMENT LEAD TIME

START

An engineer checks a change into version control



END

The change runs in production, giving value and feedback

DESIGN AND DEVELOPMENT

Like Lean product development: highly variable and uncertain

TEST, DEPLOY, OPERATIONS

Like Lean manufacturing: predictable, with short lead times and near-zero defects

Source: Kim et al. (2021), ch. 1

LEAD TIME VS. PROCESSING TIME

LEAD TIME

The clock starts when the request is made and ends when it is fulfilled.

It includes time spent waiting in queue. It is what the customer experiences.

PROCESSING TIME

The clock starts only when work begins on the request. Also called touch time or task time.

It leaves out time the work is waiting in queue.

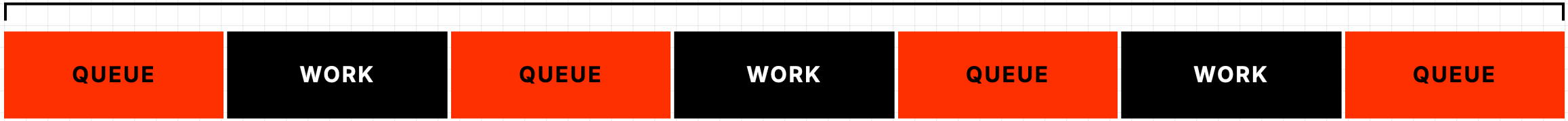
Processing time as a share of lead time is a key efficiency measure. Fast flow almost always means less waiting in queues.

Source: Kim et al. (2021), ch. 1

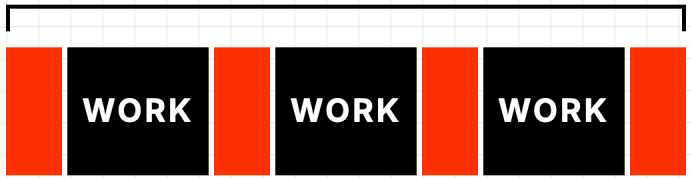
SAME WORK, SHORTER LEAD TIME

Waiting in queue Hands-on work (processing time)

LONG QUEUES: LEAD TIME IS MOSTLY WAITING



SAME WORK, SHORT QUEUES: LEAD TIME SHRINKS



Work is now about two-thirds of lead time, up from about one-fifth.

Illustration by the author, based on Fig. 1.1 in Kim et al. (2021). Proportions are illustrative.

COMMON SCENARIO: MONTHS

WHAT CAUSES IT

- Tightly coupled, monolithic systems
- Scarce integration test environments
- Long test and production environment lead times
- Heavy reliance on manual testing
- Multiple required approval processes

3 MONTHS

Example value stream, Fig. 1.2 (Edwards, 2015)

- Heroics are required at almost every stage
- At the end, nothing works once all the changes merge
- Days or weeks to find who broke it and why

Source: Kim et al. (2021), ch. 1; Fig. 1.2 from Edwards (2015)

THE DEVOPS IDEAL: MINUTES



WHAT MAKES IT POSSIBLE

A modular, loosely coupled architecture. Small teams work with high autonomy, and failures stay small and contained.

MINUTES

Deployment lead time is measured in minutes, or in the worst case, hours.

Source: Kim et al. (2021), ch. 1

SIDE BY SIDE

Area	Common scenario	DevOps ideal
Lead time	Months	Minutes (hours at worst)
Architecture	Tightly coupled monolith	Modular, loosely coupled
Testing	Manual, scarce environments	Automated and exploratory
Integration	One big merge near the end	Small changes, checked in often
Feedback	Late, fixed with heroics	Fast and constant

Source: Kim et al. (2021), ch. 1

WHY THESE MEASURES MATTER

01

CUSTOMER'S CLOCK

Lead time is what the customer feels, so we improve there first.

02

ATTACK THE QUEUES

Short lead times almost always mean less time waiting.

03

QUALITY IN THE FLOW

%C/A is how often downstream gets work usable as is.

Industry research uses the same idea. DORA defines change lead time as the time for a change to go from committed to version control to deployed in production (DORA, 2026).

Sources: Kim et al. (2021), ch. 1; DORA (2026)

REFERENCES

DORA. (2026, January 5). *DORA's software delivery performance metrics*. https://dora.dev/guides/dora-metrics/

Kim, G., Humble, J., Debois, P., Willis, J., & Forsgren, N. (2021). *The DevOps handbook: How to create world-class agility, reliability, & security in technology organizations* (2nd ed.). IT Revolution Press.

The Martin and Osterling quote and the Edwards (2015) example are cited as they appear in Kim et al. (2021). The diagram on slide 5 is an original illustration based on the idea in Fig. 1.1.